# More Trade, More Disputes?

## Measuring Alliance Effects on WTO Conflict

Peng-Ting Kuo[1] Hsuan-Wei Lee[2]

[1]Department of Government
University of Texas at Austin

[2]College of Health
Lehigh University

April 24, 2026

1. Introduction
2. Data
3. Model & Results

1 Introduction

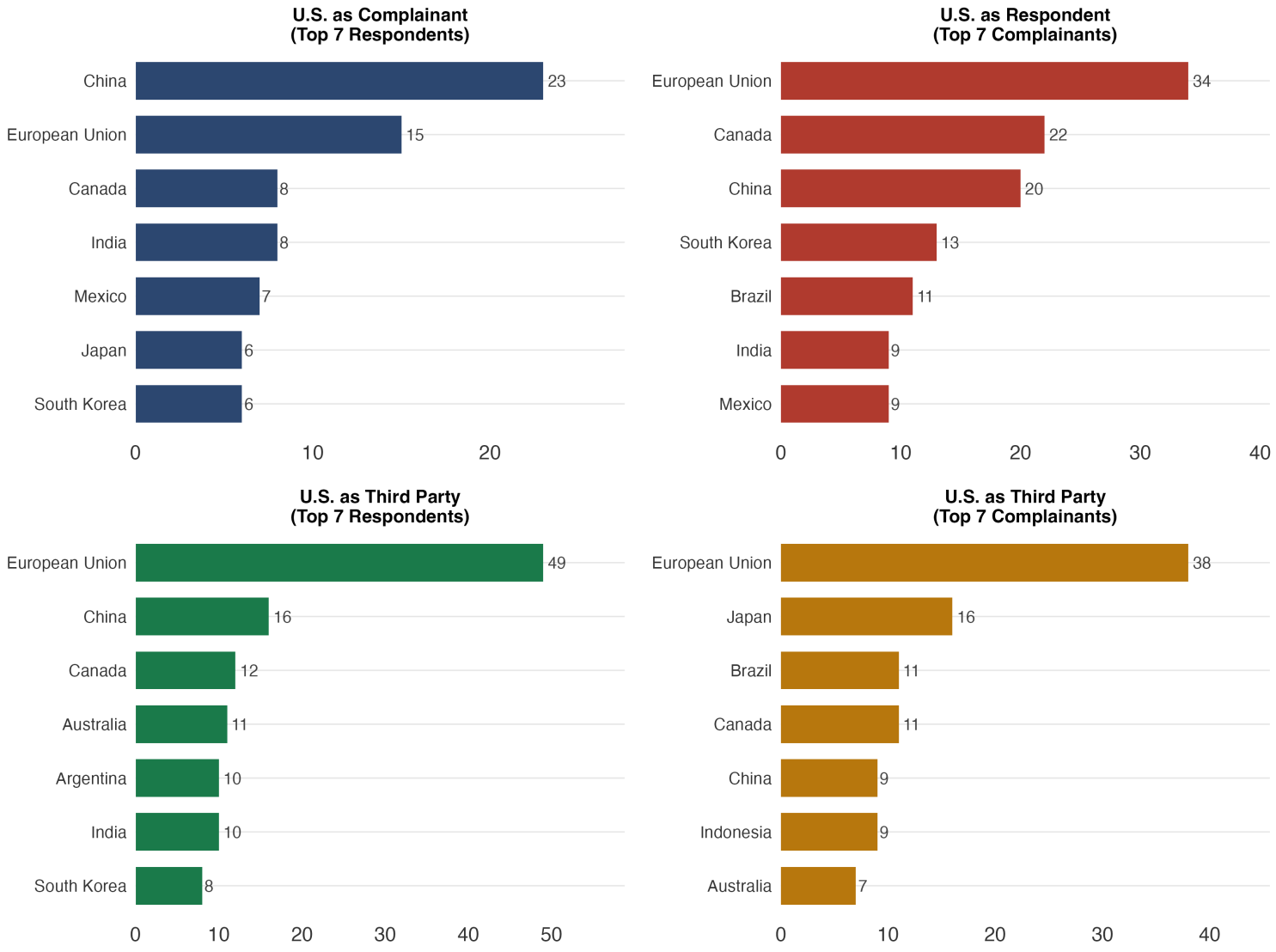
2 Data

3 Model & Results

# Puzzle

Why do allies choose the costly formal adjudication rather than the efficient bilateral diplomatic bargaining when trade conflicts arise?

# Example


U.S. as Complainant
(Top 7 Respondents)
China 23
European Union 15
Canada 8
India 8
Mexico 7
Japan 6
South Korea 6
0 10 20
U.S. as Respondent
(Top 7 Complainants)
European Union 34
Canada 22
China 20
South Korea 13
Brazil 11
India 9
Mexico 9
0 10 20 30 40
U.S. as Third Party
(Top 7 Respondents)
European Union 49
China 16
Canada 12
Australia 11
Argentina 10
India 10
South Korea 8
0 10 20 30 40 50
U.S. as Third Party
(Top 7 Complainants)
European Union 38
Japan 16
Brazil 11
Canada 11
China 9
Indonesia 9
Australia 7
0 10 20 30 40

# Why Do Allies Choose Formal Adjudication?

- Rational calculation: filing scales with economic stakes (Guzman and Simmons 2005; Busch 2007)

# Why Do Allies Choose Formal Adjudication?

- Rational calculation: filing scales with economic stakes (Guzman and Simmons 2005; Busch 2007)
- Domestic audience costs: democratic governments go public (Allee and Huth 2006; Davis 2012)

# Why Do Allies Choose Formal Adjudication?

- Rational calculation: filing scales with economic stakes (Guzman and Simmons 2005; Busch 2007)
- Domestic audience costs: democratic governments go public (Allee and Huth 2006; Davis 2012)
- Selection bias: ally disputes are too difficult for diplomacy (Gray and Potter 2020)

# Why Do Allies Choose Formal Adjudication?

- Rational calculation: filing scales with economic stakes (Guzman and Simmons 2005; Busch 2007)
- Domestic audience costs: democratic governments go public (Allee and Huth 2006; Davis 2012)
- Selection bias: ally disputes are too difficult for diplomacy (Gray and Potter 2020)
- Institutional cover: WTO absorbs domestic pressure (Rosendorff and Milner 2001)

# Why Do Allies Choose Formal Adjudication?

- Rational calculation: filing scales with economic stakes (Guzman and Simmons 2005; Busch 2007)
- Domestic audience costs: democratic governments go public (Allee and Huth 2006; Davis 2012)
- Selection bias: ally disputes are too difficult for diplomacy (Gray and Potter 2020)
- Institutional cover: WTO absorbs domestic pressure (Rosendorff and Milner 2001)

**The Remaining Puzzle**

All four predict allies should file less or only at high stakes, yet allies file more, even at low stakes, and settle early

# An International Signaling Mechanism

### The Verification Problem

Bilateral diplomacy cannot distinguish genuine enforcement from capitulation

→ No external audience
→ No reputational stakes

### WTO Solves This

Public + Irrevocable + Multilaterally Witnessed

→ Filing itself is the signal
→ Reputation achieved
→ Early settlement is rational

**Counterfactual Prediction**

If filing were purely for economic redress:
disputes should concentrate at high stakes, not low stakes

# Hypothesis

- Allied dyads initiate WTO disputes at higher rates than non-allied dyads.

# Hypothesis

- Allied dyads initiate WTO disputes at higher rates than non-allied dyads.
- Allied disputes are no more likely than non-allied disputes to escalate further on the WTO

# Hypothesis

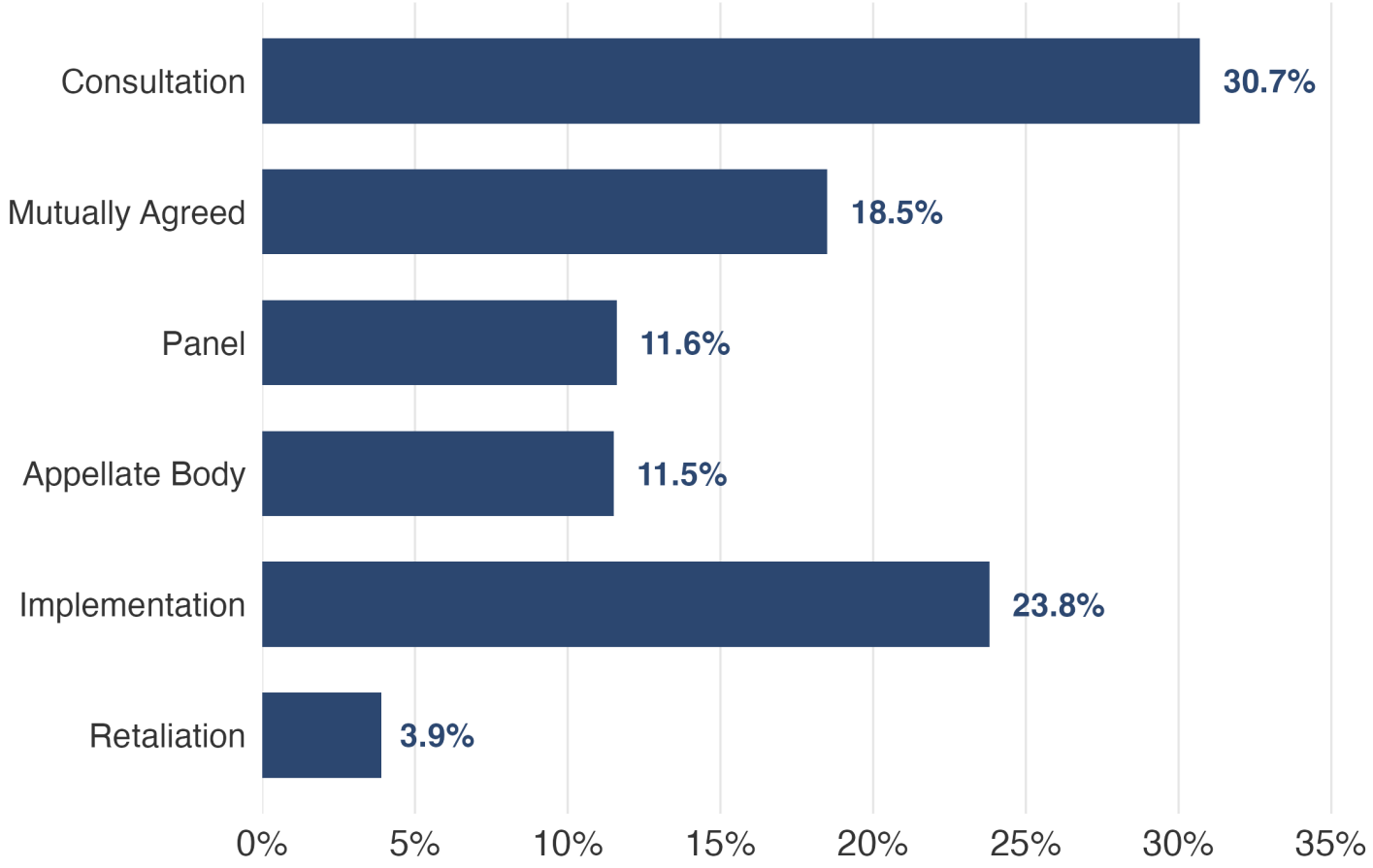
- Allied dyads initiate WTO disputes at higher rates than non-allied dyads.
- Allied disputes are no more likely than non-allied disputes to escalate further on the WTO
- Allied complainants are more likely than non-allied complainants to initiate disputes at lower levels of overall bilateral trade

1 Introduction

2 Data

3 Model & Results

# Distribution of Case Highest Stage



1 Introduction

2 Data

3 Model & Results

# TERGM Evidence: Allies File More(H1)

| | (1) No-EUN | (2) Base | (3) +UN | (4) +Ally | (5) +Demo |
|---|---|---|---|---|---|
| Edges | $8.973^*$ | $4.449$ | $6.948^*$ | $5.092$ | $5.205$ |
| | $[3.05, 15.54]$ | $[-3.00, 12.37]$ | $[0.24, 14.25]$ | $[-1.41, 11.85]$ | $[-1.98, 12.79]$ |
| Reciprocity | $1.338^*$ | $1.446^*$ | $1.411^*$ | $1.423^*$ | $1.405^*$ |
| | $[0.50, 2.14]$ | $[0.74, 2.08]$ | $[0.66, 2.09]$ | $[0.75, 2.05]$ | $[0.64, 2.09]$ |
| GW In-degree | $-2.342^*$ | $-2.126^*$ | $-2.072^*$ | $-2.106^*$ | $-2.095^*$ |
| | $[-3.09, -1.41]$ | $[-2.84, -1.18]$ | $[-2.77, -1.13]$ | $[-2.78, -1.15]$ | $[-2.78, -1.24]$ |
| Bilateral Trade | $0.445^*$ | $0.389^*$ | $0.429^*$ | $0.395^*$ | $0.428^*$ |
| | $[0.36, 0.56]$ | $[0.29, 0.52]$ | $[0.33, 0.54]$ | $[0.31, 0.53]$ | $[0.31, 0.57]$ |
| Alliance | $0.518^*$ | $0.417^*$ | | $0.360^*$ | $0.394^*$ |
| | $[0.25, 0.82]$ | $[0.25, 0.60]$ | | $[0.21, 0.55]$ | $[0.23, 0.58]$ |
| UN Voting Dist. | | | $0.299^*$ | $0.251^*$ | $0.290^*$ |
| | | | $[0.04, 0.49]$ | $[0.01, 0.45]$ | $[0.03, 0.51]$ |
| Resp. Democracy | | | | | $1.078^*$ |
| | | | | | $[0.13, 1.83]$ |
| Controls | Included | Included | Included | Included | Included |
| Num. obs. | 756,378 | 765,774 | 756,378 | 756,378 | 665,608 |

*Notes:* 95% CIs in brackets. $^*p < 0.05$. Controls include standard dyadic and network terms.

# Ordinal Logit: Once Filed, Alliance Politics Ends (H2)

| | (1) Baseline | (2) +Node | (3) +UN | (4) +Severity |
|---|---|---|---|---|
| Alliance | 0.024 | −0.043 | −0.019 | −0.016 |
| | (0.118) | (0.205) | (0.275) | (0.269) |
| Bilateral Trade | 0.144*** | 0.107** | 0.135** | 0.128* |
| | (0.024) | (0.039) | (0.052) | (0.051) |
| GDPPC Gap | | 0.082 | 0.127 | 0.134 |
| | | (0.093) | (0.137) | (0.136) |
| UN Voting Dist. | | | −0.012 | −0.007 |
| | | | (0.132) | (0.132) |
| Dispute Severity | | | | 0.224 |
| | | | | (0.172) |
| N Third Parties | | 3.496** | 3.465** | 3.445** |
| | | (1.063) | (1.076) | (1.056) |
| Controls | No | Included | Included | Included |
| Num. obs. | 668 | 666 | 654 | 649 |

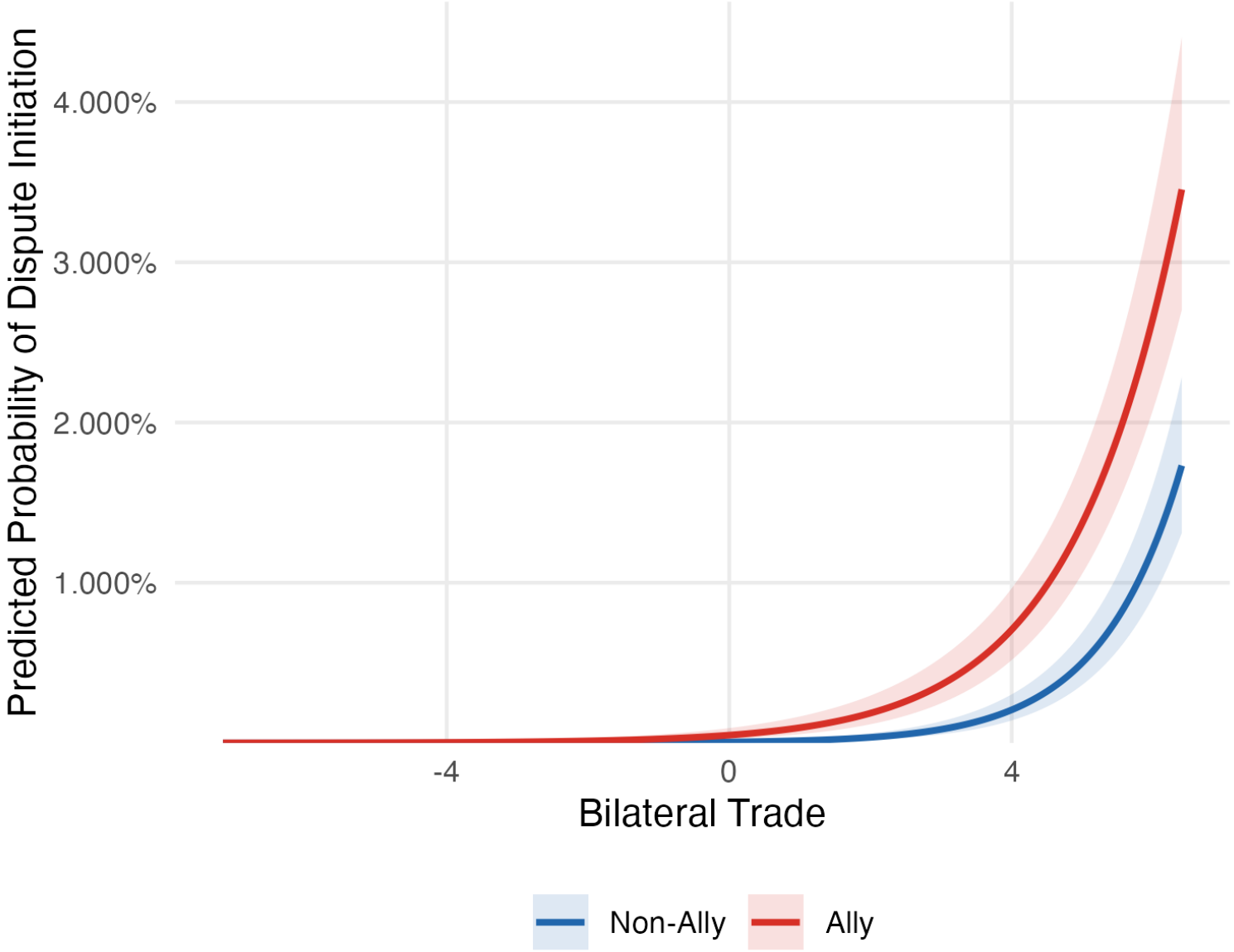
*Notes:* $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Cluster-robust SEs in parentheses.

# Logit: Alliance and WTO Dispute Initiation (H3)

| | (1) Baseline | (2) +Interact | (3) +DemInteract |
|---|---|---|---|
| Alliance | 0.406** | 2.046*** | 2.549*** |
| | (0.152) | (0.427) | (0.521) |
| Bilateral Trade | 0.785*** | 0.894*** | 0.889*** |
| | (0.045) | (0.050) | (0.052) |
| Democracy | | | 1.292*** |
| | | | (0.359) |
| **Alliance × Trade** | | **-0.223*** ** | **-0.219*** ** |
| | | (0.057) | (0.058) |
| Alliance × Democracy | | | −0.745 |
| | | | (0.465) |
| UN Voting Dist. | 0.143*** | 0.214** | 0.220** |
| | (0.081) | (0.083) | (0.084) |
| Comp. Democracy | 1.075** | 1.122*** | 1.486*** |
| | (0.336) | (0.340) | (0.356) |
| Controls | Included | Included | Included |
| Num. obs. | 482,950 | 482,950 | 482,950 |

*Notes:* $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$. Cluster-robust SEs in parentheses.

# Marginal Effect on Alliance Interaction Term



# Take Away

- Formal alliance ties independently predict WTO dispute initiation, even after conditioning on trade volume and political alignment

# Take Away

- Formal alliance ties independently predict WTO dispute initiation, even after conditioning on trade volume and political alignment
- Alliance status has no effect on procedural escalation once filed, WTO institutional logic (third-party dynamics) takes over

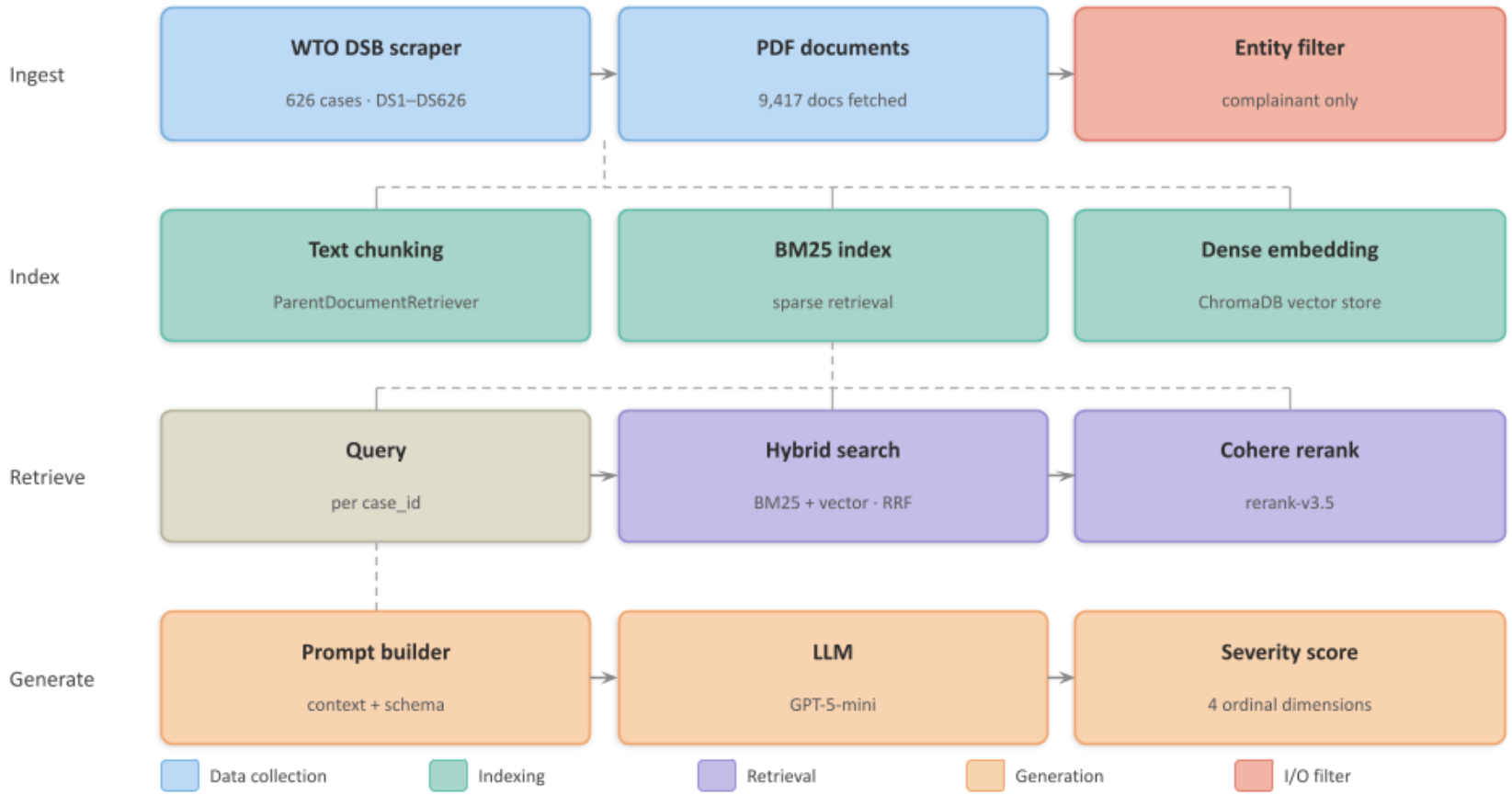
# Take Away

- Formal alliance ties independently predict WTO dispute initiation, even after conditioning on trade volume and political alignment
- Alliance status has no effect on procedural escalation once filed, WTO institutional logic (third-party dynamics) takes over
- Allies file even when the overall bilateral trade relationship is small

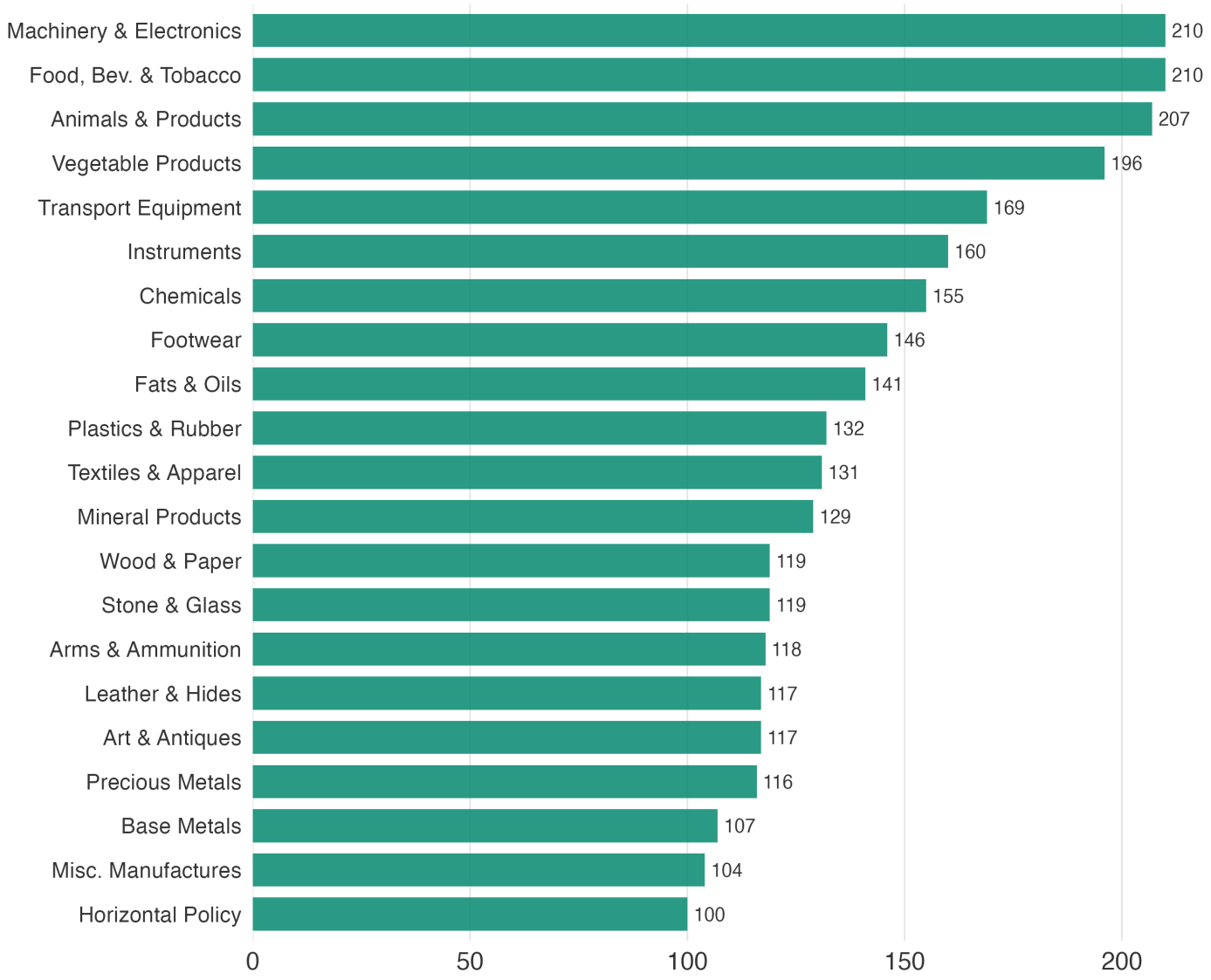
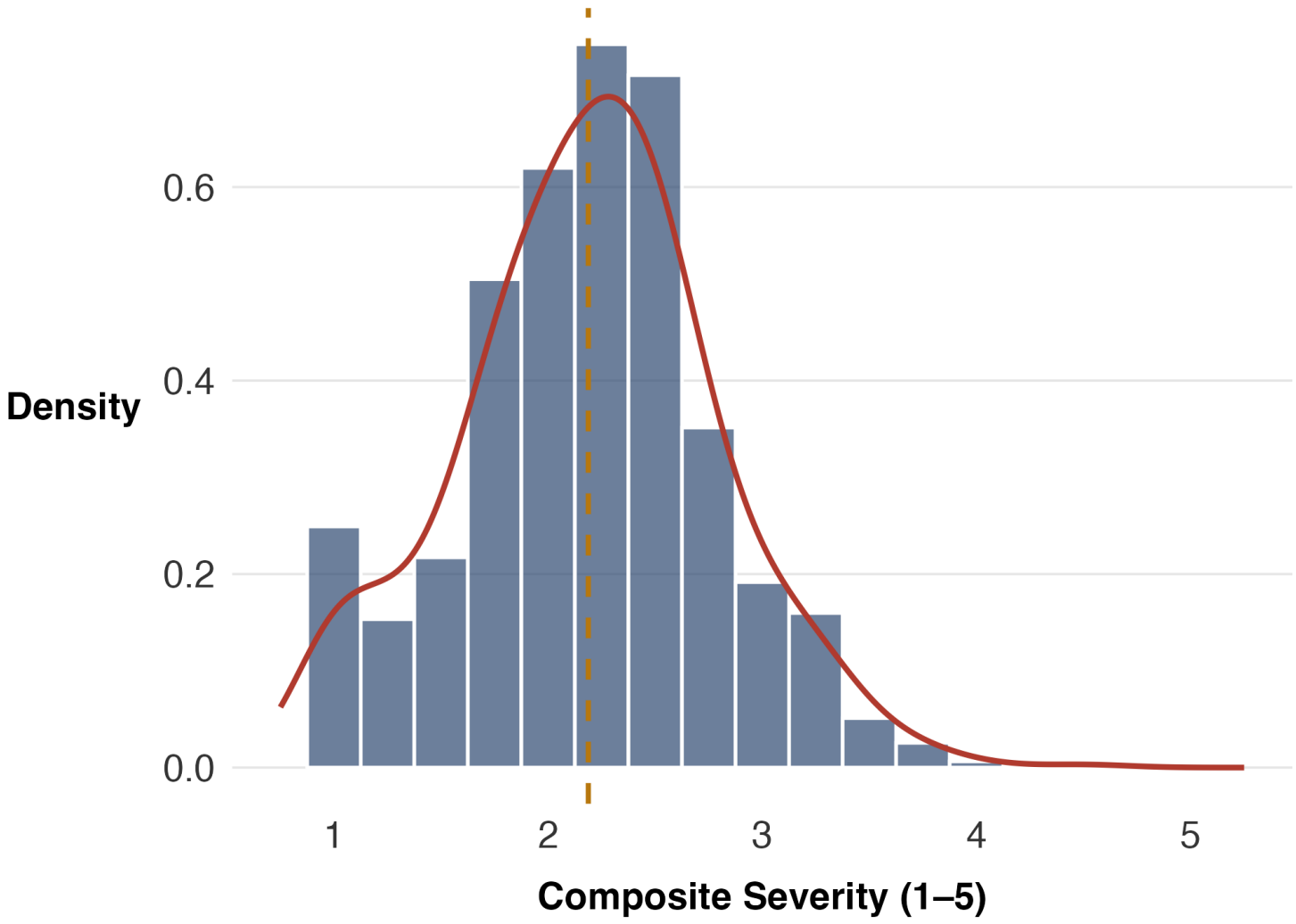
# Future Work

- RAG infrastructure
- Measurement of rhetoric severity
- Measure the section volume in each case

# RAG Architecture



# WTO Dispute Trade Section



# Severity Score Distribution